Test Plan

Prerequisites

```
docker compose up --build
# Wait ~60s for Brain to start (downloads Sivas OSMnx map on first run)
# Open http://localhost:8000
```

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1. Service Health Checks

Check	Expected
`curl http://localhost:3000/health`	`{"status":"ok"}`
`curl http://localhost:8001/api/health`	`{"status":"online","service":"SMART LOGISTICS: The Brain"}`
`curl http://localhost:7777/health`	`{"status":"ok","service":"TomTom Mock API"}`
`docker logs smart_logistics_brain`	`✅ Successfully seeded N physical street segments.`

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2. Authentication

- Navigate to `http://localhost:8000`
- Login: `johndoe@smartlogistics.com` / `password123`
- **Expected:** redirected to courier dashboard, no blank screen
- **Failure mode to check:** if `isAuthenticated` doesn't flip to `true`, the JWT decode failed

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3. Manifest Load & Sidebar Cards

Immediately after login:

- **3 red pin markers** appear on the Sivas map (around 39.75, 37.01)
- Right sidebar shows **3 delivery cards** with:
  - Client first name: **Ali**, **Ayse**, **Mehmet**
  - Phone: `+90 346 111 2233`, `+90 346 222 3344`, `+90 346 333 4455`
  - Time range: `09:00 – 11:00`, `11:30 – 13:00`, `14:00 – 16:00`
  - Street name from PostGIS (e.g. a Turkish road name — if segments table is empty, falls back to `39.7500, 37.0150`)
- **Browser console:** `[HEALTH CHECK] Sent 3 stops to courier frontend`

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4. Initial Brain Health Check → Blue Route Line

Within ~5–10 seconds of login:

- **Blue polyline** appears on the map connecting the stops

- **Gateway logs:** `[HEALTH CHECK] Pushed to Brain for DRV-884`
- **Brain logs:** `Published CONTINUE for MANIFEST-TODAY` (or `RE-ROUTE` / `DELAY_DEPARTURE` if HEAVY traffic)
- If an **ActionCard** appears on the left panel, the full AI pipeline is confirmed working

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5. Courier Simulator Dot — T=0 Position

After ~30 seconds (container sleep):

- **Green dot** appears at depot `(39.720, 37.010)` — visibly **south** of all delivery pins, not overlapping them
- Click the dot → popup shows `Vehicle ID: DRV-884 | Status: ON-TIME | Speed: 0 km/h`
- **Gateway logs:** `[Courier Simulator] Unit DRV-884 online.`

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6. Courier Movement — Telemetry Pings

Every 30 seconds:

- Green dot moves **northward** toward Stop 1 (Ali, 39.750)
- Each step ≈ 188 m; movement should be clearly visible across pings
- **Gateway logs:** `[Courier] Ping N: 39.72XXX, 37.0XXXX | XX.X km/h → Ali`

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7. Predictive Traffic Alert → ActionCard

On ping 2 (T≈90s, heading established):

- **Gateway logs:** `[TRAFFIC ALERT] 5km-ahead congestion for DRV-884 (high)`
- **Brain logs:** `Published REQUEST_ALTERNATE_PATH for MANIFEST-TODAY`
- **Left panel:** ActionCard appears with:
  - Severity badge: `REROUTE` or `INFO`
  - Reason: "*Sequence remains optimal but physical path blocked. Requesting alternate route.*"
  - Impact grid: route health + time saved
  - **Approve Swap** and **Refuse** buttons

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8. Approve Route

Click **Approve Swap**:

- Button → `Syncing...` → `Applied` → card **auto-removes** after 3s
- **Gateway logs:** `APPROVAL_SUCCESS`
- **DB check:** `SELECT status FROM daily_manifest WHERE manifest_id = 'MANIFEST-TODAY';` → `IN_TRANSIT`
- **DB check:** `SELECT delivery_order FROM manifest_stops ORDER BY delivery_order;` → reordered if Brain suggested swap

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9. Refuse Route

On a subsequent recommendation, click **Refuse**:

- Button triggers `REFUSE_ROUTE` WebSocket message
- Card removes immediately (status goes to `Syncing...` then gets `REFUSE_ROUTE_ACK`)
- **No DB change** — manifest stays unchanged

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10. Delivery Completion — Grey Stop

After ~4–5 minutes (courier reaches Stop 1):

- **Gateway logs:** `[DELIVERY] Stop 1 marked DELIVERED for DRV-884`
- **Sidebar card for Ali:** turns grey, label changes to `COMPLETED`, MapPin dims
- **Map pin for Ali:** becomes semi-transparent grey pin (opacity 0.45)
- Click grey pin → popup shows `Ali ✓ Delivered | <street name> | +90 346 111 2233`
- **DB check:** `SELECT delivery_status FROM manifest_stops WHERE stop_id = 1;` → `DELIVERED`
- Courier dot continues toward Stop 2 (Ayse)

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11. Map Popup Verification

Click any active (red) stop pin:

- Shows client first name + street name + phone number
- No raw lat/lon text visible to the user

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12. Break Mode

Click **TAKE BREAK**:

- Sidebar dims, shows ☕ "On Break" overlay
- Left alerts panel hides (no ActionCards visible during break)
- Click **END BREAK** → returns to normal

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13. Logout & Re-login

Click **LOGOUT**:

- Redirected to login screen, token cleared from localStorage
- Re-login → manifest reloads fresh, `GET_DAILY_MANIFEST` fires again
- If Stop 1 is marked DELIVERED in DB, it should **not appear** in the sidebar (query filters `status = 'PENDING'`)

Quick DB Verification Commands

```
docker exec -it smart_logistics_db psql -U postgres -d smart_logistics -c \
  "SELECT stop_id, delivery_status FROM manifest_stops;"

docker exec -it smart_logistics_db psql -U postgres -d smart_logistics -c \
  "SELECT status FROM daily_manifest WHERE manifest_id = 'MANIFEST-TODAY';"

docker exec -it smart_logistics_db psql -U postgres -d smart_logistics -c \
  "SELECT COUNT(*) FROM segments;"
```